

Final Exam: I3307E

Duration: 1h30

3rd Year – 2nd semester 2025 – 2nd session

Documents: non authorized

Date: 20/07/2025

FINAL : I3307F

Durée : 1h30

Lima

3^{ème} année – 2^{ème} semestre 2025 – 2^{ème} session

Documents non permis

Date : 20/07/2025

Exercise I

(20 pts)

- Consider the languages $L_1 = \emptyset$ and $L_2 = \{a\}$. Calculate $L_1 L_2^* \cup L_1^*$?
Considérons les langages $L_1 = \emptyset$ et $L_2 = \{a\}$. Calculer $L_1 L_2^* \cup L_1^*$?
- Let $L = \{ab, aa, baa\}$. Which of the following strings are in L^* : $abaabaaabaa$, $aaaabaaaa$, $baaaaabaaaab$, $baaaaabaa$? Which strings are in L^4 ? Justify.
Soit $L = \{ab, aa, baa\}$. Lesquels des mots suivants se trouvent dans L^* : $abaabaaabaa$, $aaaabaaaa$, $baaaaabaaaab$, $baaaaabaa$? Quels mots sont dans L^4 ? Justifier.
- Are there languages for which $\bar{L}^* = (\bar{L})^*$? Justify.
Y a-t-il des langages pour lesquelles $\bar{L}^* = (\bar{L})^*$? Justifier.
- Let L be the language corresponding to the RE $(a + aab + aaab)^*(aa + aaa)$. Which of the following strings is in L ? $aaaabaa$, $aaabbbaa$, $aaabaaba$, $aababaabaa$. Justify.
Soit L le langage correspondant à l'ER $(a + aab + aaab)^*(aa + aaa)$. Laquelle des mots suivants est en L ? $aaaabaa$, $aaabbbaa$, $aaabaaba$, $aababaabaa$. Justifier.

Exercise II

(15 pts)

Design a DFA that recognizes the following languages:

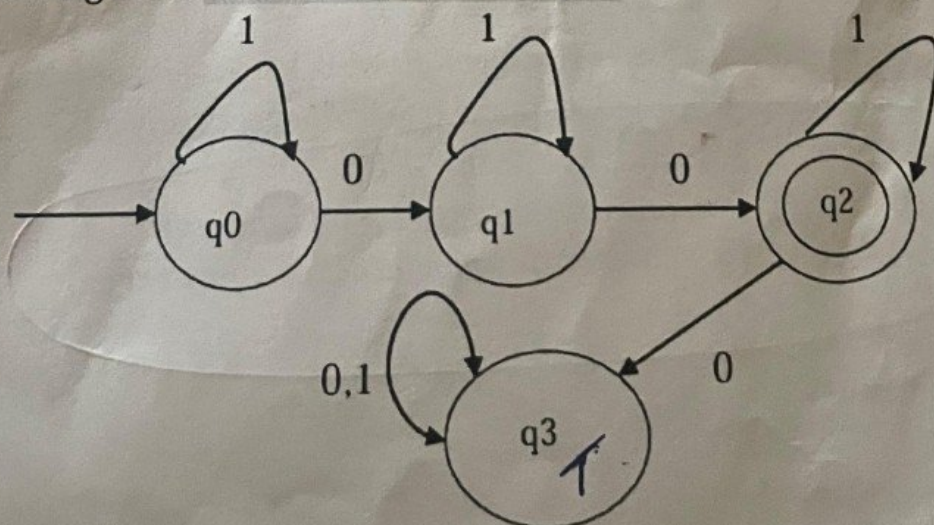
Donnez un DFA qui reconnaît les langages suivants :

- $L = \{w \in \{a, b\}^* \mid w \text{ contains at least one } a \text{ and exactly two } b's\}$
 $L = \{w \in \{a, b\}^* \mid w \text{ contient au moins un } a \text{ et exactement deux } b\}$
- $L = \{w \in \{a, b\}^* \mid w = ba^n : n \geq 1, n \neq 5\}$
- $L = \{w \in \{a, b\}^* \mid |w|_a \bmod 3 > |w|_b \bmod 3\}$

Exercise III

(15 pts)

Consider the following DFA: Considérons l'AFD suivant :



- Provide a formal description of the DFA. Donnez une description formelle de l'AFD.

Ex 1

a) $L_1 = \emptyset$ $L_2 = \{a\}$

$$L_1 L_2^* \cup L_1^* = \emptyset \cdot \{a\}^* \cup L_1^* = \emptyset \cup \{\epsilon\} = \{\epsilon\}$$

b) $L = \{ab, aa, baa\}$

$$abaa baa abaa \in L^5 \subseteq L^*$$

$$aaaa baa aa \in L^4 \subseteq L^*$$

$$baa aa abaa ab \notin L^*$$

$$baa aa abaa \in L^4 \subseteq L^*$$

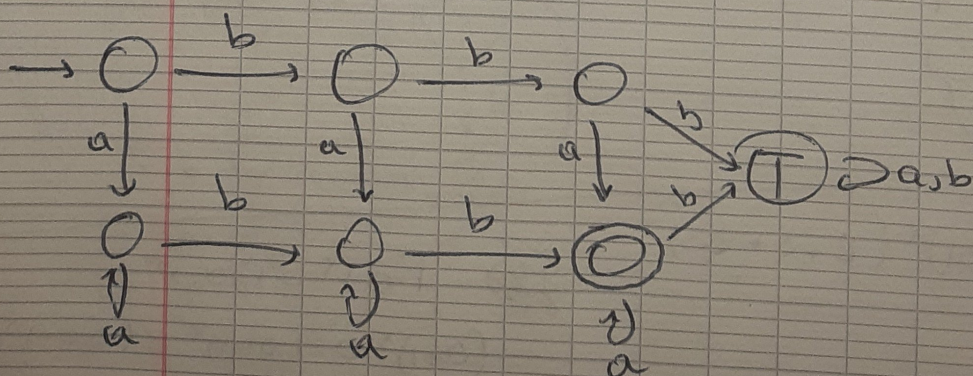
c) No, L^* and $(\bar{L})^*$ both contains ϵ
since $\bar{L}^*$ doesn't contain ϵ
 (L^*) and $(\bar{L})^*$ can't be equal

d) $(a+ab+aaab)^*(aa+aaaa)$

every string in this language must end with aa or $aaaa$
and every b must be preceded at least 2 a 's
so "aaabaaba" and "aababaaabaa" are not accepted

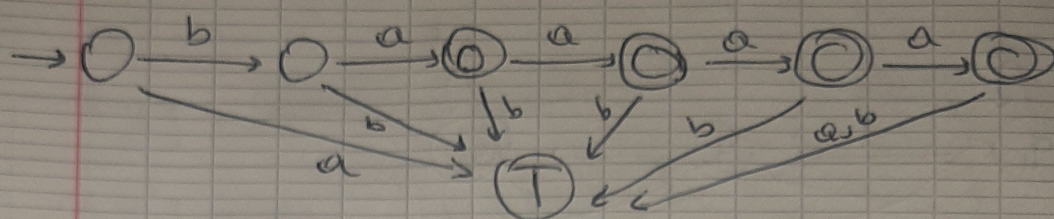
Ex 2

a) at least one a : zero a , one or more a
exactly two b : zero b , one b , two b , ≥ 2 b (trap)
 $2 \times 3 = 6$ states + 1 trap state



Ex 2

b) $w = ba^n$; $n > 0$ & $n \neq 5$



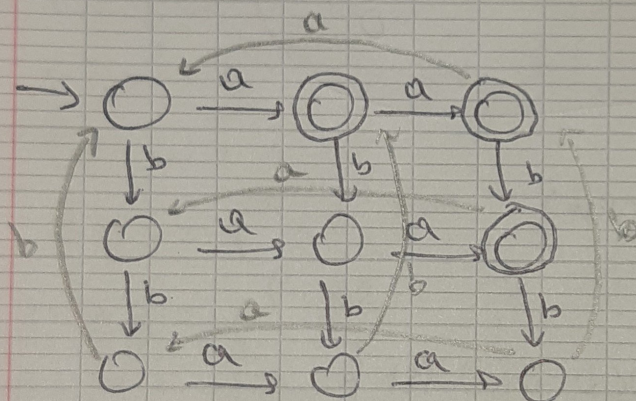
c) $|w|_a \bmod 3 \in \{0, 1, 2\}$

$|w|_b \bmod 3 \in \{0, 1, 2\}$

$3 \times 3 = 9$ states

$|w|_a > |w|_b \Rightarrow (|w|_a, |w|_b) \in \{(1, 0), (2, 0), (2, 1)\}$

3 accepting states



Ex 3

a) DFA = $\{Q, \Sigma, s, F, \delta\}$

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

$s = q_0$

$F = \{q_2\}$

δ = transition function table

	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_1
$^x q_2$	q_3	q_2
q_3	q_3	q_3

b) $L = \{w \in \{0, 1\}^* \mid |w|_0 = 2\}$

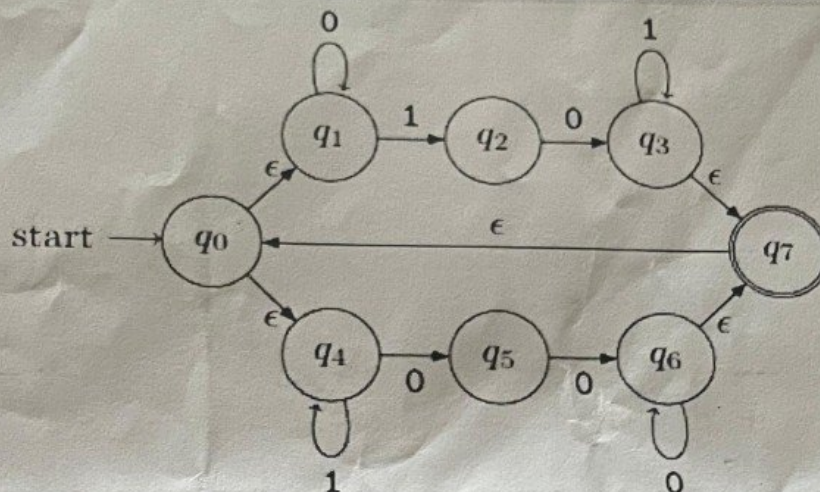
c) $1^* 0 1^* 0 1^*$

- b. What is the language accepted by this DFA. Quelle est le langage accepté par ce AFD ?
 c. Convert the DFA into a regular expression. Convertir l'AFD en une expression régulière.

Exercise IV

(15 pts)

Consider the following ϵ -NFA, give the corresponding NFA by eliminating ϵ -transitions.
 Soit l' ϵ -AFN suivant, donnez l'AFN correspondant en éliminant les transitions par ϵ .



Exercise V

(20 pts)

Consider the alphabet $\Sigma = \{a, b\}$. Give a regular expression for each of the following languages:
 Considérons l'alphabet $\Sigma = \{a, b\}$. Donnez une expression régulière pour chacun des langages suivants :

- a. $L_1 = \{a^n b^m : n \geq 3, m \text{ is even}\}$. $L_1 = \{a^n b^m : n \geq 3, m \text{ est pair}\}$. $(a^3 a^*) (b^2)^*$
 b. $L_2 = \{a^n b^m : n \geq 4, m \leq 3\}$. $L_2 = \{a^n b^m : n \geq 4, m \leq 3\}$. $(a^4 a^*) (b^0 + b^1 + b^2 + b^3)$
 c. $L_3 = \{w \mid w \text{ contains both } 000 \text{ and } 111 \text{ as substrings}\}$. $(0+1)^* (000) (111) (0+1)^*$
 $L_3 = \{w \mid w \text{ contient à la fois } 000 \text{ et } 111 \text{ comme sous-chaînes}\}$.
 d. $L_4 = \{w \mid |w| \text{ is even and contains } 010 \text{ as substring}\}$. $(0+1)^* 010 (0+1)^*$
 $L_4 = \{w \mid |w| \text{ est paire et contient } 010 \text{ comme sous-chaîne}\}$.

Exercise VI

(15 pts)

Using the Pumping Lemma, prove that the following languages are not regular:
 En utilisant le lemme de pompage, montrez que les langages suivants ne sont pas réguliers :

- a. $L_1 = \{a^i b^k c^m d^r \mid i \geq m \geq 0, k \geq 0, r \geq 0\}$.
 b. $L_2 = \{w \in \{(,)\}^* \mid w \text{ contains balanced parentheses}\}$.
 $L_2 = \{w \in \{(,)\}^* \mid w \text{ contient des parenthèses équilibrées}\}$

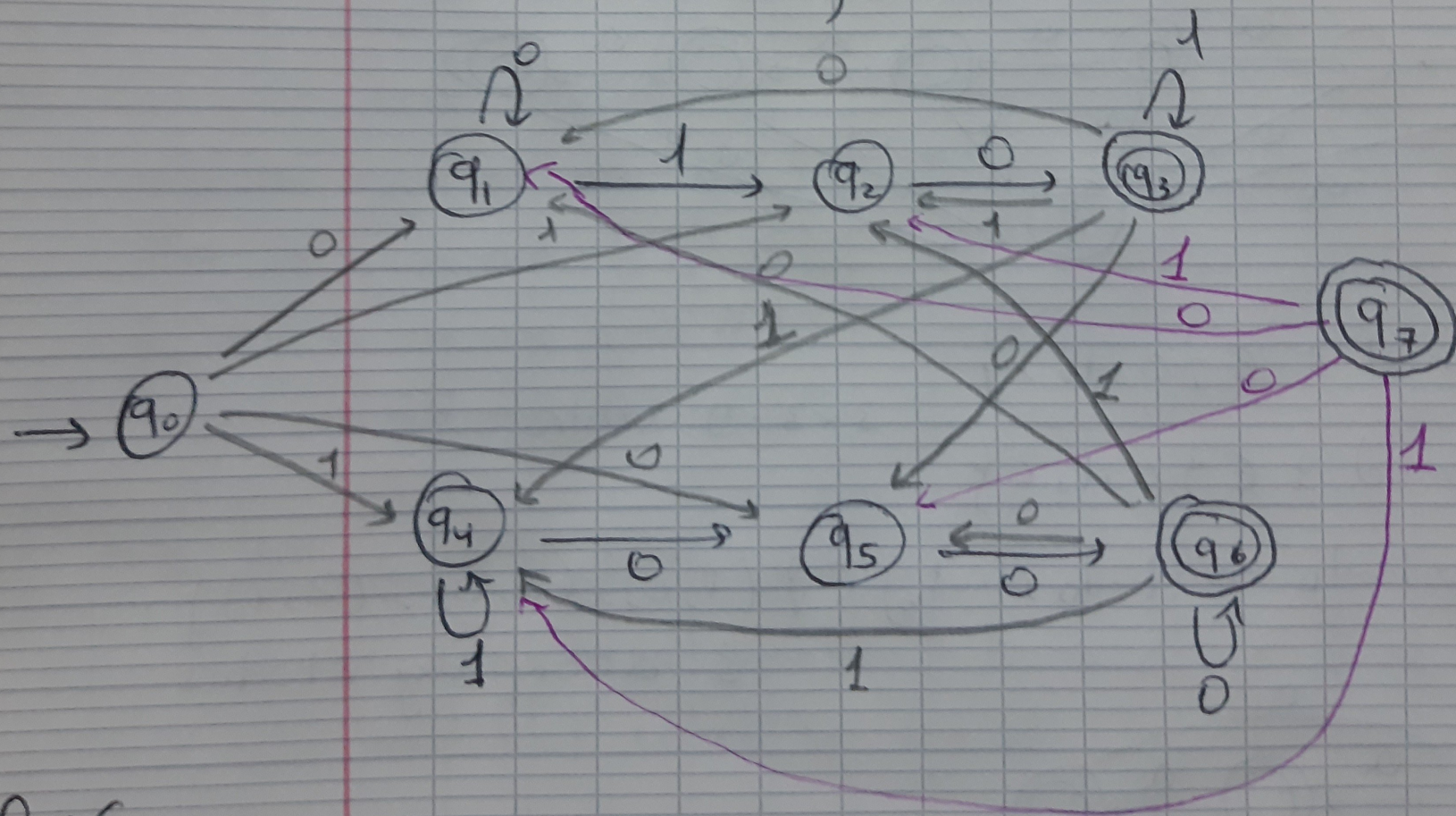
Ex 4

$$q_0 = \{q_0, q_1, q_4\}$$

$$q_3 = \{q_3, \underline{q_7}, q_0, q_1, q_4\}$$

$$q_6 = \{q_6, \underline{q_7}, q_0, q_1, q_4\}$$

$$q_7 = \{\underline{q_7}, q_0, q_1, q_4\}$$



Ex 5

Qx5

a) $L = \{a^n b^m; n \geq 3 \text{ and } m \text{ is even}\}$

$$aaa(a)^x(bb)^x$$

c) $L = \{w \mid w \text{ contains both } 000 \text{ and } 111\}$

$$(0+1)^x 000 (0+1)^x 111 (0+1)^x + (0+1)^x 111 (0+1)^x 000 (0+1)^x$$

b) $L = \{a^n b^m; n \geq 4; m \leq 3\}$

$$aaaa(a)^x (\epsilon + b + bb + bbb)$$

d) $L = \{w \mid w \text{ is even and contains } 010\}$

Note: 010 is odd, so we need odd characters so $|w|$ becomes even

$$\text{odd characters} = ((0+1)(0+1))^x (0+1)$$

$$(((0+1)(0+1))^x (0+1) 010 ((0+1)(0+1))^x (0+1))$$

$$+ (((0+1)(0+1))^x 010 (((0+1)(0+1))^x (0+1))$$

ex 6

a) $L = \{ a^i b^k c^m d^r : i \geq m \}$

let $w = a^p c^p \in L$

$xyz = \cancel{a^p b^k c^p} a^\alpha a^\beta a^{p-\alpha-\beta} b^p$

$xy^i z = a^{p+\beta(i-1)} c^p$

~~let $i=2$, $xy^2 z = a$~~

let $i=0$, $xy^0 z = a^{p-\beta} c^p \notin L$

$\beta > 0$ and $p-\beta < p \Rightarrow L$ not regular

b) $L = \{ w \in \{ (,) \}^* : w \text{ contains balanced parentheses} \}$

assume L is regular

let $w = ({}^p)_p$

$xyz = ({}^\alpha ({}^\beta ({}^{p-\alpha-\beta})_p$

$xy^i z = ({}^{p+\beta(i-1)})_p$

$i=2$, $xy^2 z = ({}^{p+\beta})_p \notin L$

$p+\beta > p$

L not regular.